NexusRAG: What Actually Improves Local Retrieval for Scientific Claims, and How to Verify the Answer

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Abstract

Local retrieval-augmented generation tools for scientific reading routinely ship a hybrid retriever, a reranker, and a "verifier," but almost never report which of these components actually helps. We treat the question empirically. On two BEIR benchmarks—SciFact (300 claims, 5183 abstracts) and NFCorpus (323 queries, 3633 documents)—we run a strictly additive ablation with exact search, bootstrap confidence intervals, paired randomization tests, and Holm correction, entirely on CPU. The dominant factor is the embedding model: replacing the common all-MiniLM-L6-v2 with bge-small-en-v1.5 moves dense retrieval from below BM25 (0.648 nDCG@10) to clearly above it (0.708), a paired gain of +0.060 (p=<0.001). Reciprocal-rank fusion then beats BM25 on both corpora (p=0.002 and p=<0.001) and is the best single system on NFCorpus. We add a confidence-gated corrective re-retrieval that expands and re-queries only when the first pass is weak, and find it matches plain fusion at negligible cost while a cross-encoder reranker is far slower and no better. Finally, we recast faithfulness verification as evidence detection and find that a relevance cross-encoder—not a dedicated NLI model—is the strongest cheap verifier (0.755 vs. 0.688 ROC-AUC). Every number is regenerated from committed results by make eval on a CPU laptop. Code and data: https://github.com/urme-b/NexusRAG.

1 Introduction

The hard part of "what does the literature say about X?" is finding the few passages that bear on X. Retrieval-augmented generation (RAG) (Lewis et al., 2020) couples a retriever with a language model so answers are grounded in retrieved text, and many open-source tools apply this recipe to PDFs of papers. Two things are usually missing.

First, the retriever is rarely measured. A typical project ships a hybrid retriever—dense embeddings plus BM25, fused and reranked—without ground-truth relevance judgements, so it is impossible to say which parts help. Second, verification is often cosmetic: systems confirm that a citation marker such as [2] points to a source that exists, never that the source entails the sentence attached to it.

We take NexusRAG, a local RAG application of exactly this kind, and answer the questions its original form could not:

1. Which components earn their place? We build a strictly additive ablation and evaluate it on SciFact (Wadden et al., 2020) and NFCorpus, both via the BEIR (Thakur et al., 2021) judgements, with bootstrap CIs and paired randomization tests.

2. Does a corrective loop help, and at what cost? We add a confidence-gated pseudo-relevance-feedback (PRF) pass and compare its cost/quality against a cross-encoder reranker.

3. Can verification be made real? We replace the citation-index check with an NLI model and evaluate it as an evidence detector against lexical and cross-encoder baselines, not a zero-by-construction strawman.

Our contribution is not a new retrieval algorithm. It is an honest, component-wise measurement that isolates what matters (the embedder, then fusion), a cheap corrective alternative to reranking, and a content-inspecting verifier evaluated fairly—all CPU-only and reproducible from one command.

2 Related Work

Hybrid retrieval. Dense retrievers (Karpukhin et al., 2020; Reimers and Gurevych, 2019) and lexical scoring (Robertson and Zaragoza, 2009) have complementary failure modes, and fusing their ranked lists with reciprocal rank fusion (RRF) (Cormack et al., 2009) is a strong training-free baseline. Instruction-tuned sentence encoders such as BGE (Xiao et al., 2024) and E5 (Wang et al., 2022) sharply improved small-model dense retrieval on BEIR; our study quantifies how much of the usual pipeline that improvement subsumes. Cross-encoder reranking (Nogueira and Cho, 2019) re-scores a shortlist with full query–document attention at high cost.

Corrective and self-reflective RAG. CRAG (Yan et al., 2024) grades retrieved passages and triggers corrective actions when retrieval looks poor, and Self-RAG (Asai et al., 2024) trains a model to critique its own retrieval. Our corrective loop follows this pattern but uses a deterministic, LLM-free confidence gate and classical pseudo-relevance feedback (Lavrenko and Croft, 2001), so the result is reproducible without a generator.

Faithfulness. RAGAS (Es et al., 2024) decomposes an answer into claims and checks each against the context, usually with a large judge model. We pursue the same goal with a small local NLI cross-encoder (He et al., 2021) and, because SciFact provides sentence-level gold evidence, evaluate the grounding signal directly as a detector rather than through another model.

3 The NexusRAG System

NexusRAG ingests documents (PDF, DOCX, TXT, MD), chunks them, and indexes each chunk as both a dense vector and a BM25 entry. At query time it retrieves with each method to a fixed depth, fuses the lists with RRF (smoothing constant k=60), and—if the top dense similarity is below a threshold τ=0.55—expands the query with frequent terms from the first-pass documents and re-retrieves, fusing the two passes. A local model then answers using only the retrieved passages with inline citations, and an NLI model checks that each answer sentence is entailed by its sources. All retrieval uses exact (brute-force) cosine search, so results are deterministic.

4 Experimental Setup

Data. SciFact via BEIR is a corpus of 5183 abstracts with 300 test claims and abstract-level judgements; NFCorpus is a biomedical benchmark of 3633 documents and 323 test queries. For the faithfulness experiment we use the original SciFact release, which adds sentence-level evidence labels (188 dev claims with evidence).

Table 1: Additive ablation on SciFact (300 claims, 5183 abstracts). Best column values in bold; * marks $p < 0.05$ vs. BM25 after Holm correction.

System	nDCG@10	R@5	R@10	R@20	MRR	MAP	p vs BM25
BM25	0.666	0.719	0.790	0.821	0.636	0.626	–
Dense	0.708	0.763	0.836	0.873	0.680	0.667	0.032*
Hybrid (RRF)	0.704	0.763	0.840	0.891	0.672	0.662	0.002*
+ Adaptive weights	0.703	0.762	0.840	0.886	0.670	0.660	0.001*
+ Corrective PRF	0.703	0.762	0.840	0.886	0.670	0.660	0.001*

Table 2: Additive ablation on NFCorpus (323 queries, 3633 documents). Same conventions as Table 1.

System	nDCG@10	R@5	R@10	R@20	MRR	MAP	p vs BM25
BM25	0.311	0.120	0.150	0.172	0.518	0.136	–
Dense	0.340	0.127	0.161	0.197	0.538	0.150	0.009*
Hybrid (RRF)	0.353	0.130	0.165	0.209	0.570	0.162	<0.001*
+ Adaptive weights	0.349	0.133	0.162	0.201	0.566	0.158	<0.001*
+ Corrective PRF	0.349	0.133	0.162	0.201	0.566	0.158	<0.001*

Metrics. We report nDCG@10, Recall@$\{5, 10, 20\}$, MRR, and MAP with binary relevance, each system retrieving to depth 50. We compute 95% confidence intervals by bootstrapping over queries (10,000 resamples) and test each system against BM25 with a two-sided paired randomization test (10,000 permutations, add-one estimator), applying a Holm correction across the family of comparisons. Marked rows (*) survive Holm at α=0.05.

Models. The dense model is bge-small-en-v1.5 with the recommended query instruction; the reranker is ms-marco-MiniLM-L-6-v2; the NLI model is nli-deberta-v3-small. Everything runs on CPU; no API keys are required.

5 Results

5.1 What the components contribute

Table 1 and Table 2 report the additive ladder; Figure 1 shows nDCG@10 with bootstrap intervals.

The first row tells most of the story. With the common MiniLM encoder, dense retrieval (0.648 nDCG@10 on SciFact) is below BM25 (0.666)—the well-known reason hybrids are popular. Swapping in BGE lifts dense retrieval to 0.708, a paired improvement of +0.060 nDCG@10 (p=<0.001) and now clearly above BM25. The embedder, not the surrounding machinery, is the dominant factor.

Given a strong dense model, RRF fusion still helps where the two signals are complementary: on NFCorpus it is the best single system (0.353 vs. BM25 0.311, p=<0.001), and on SciFact it beats BM25 (0.704, p=0.002) while raising recall, even though it does not exceed the already-strong dense model on nDCG. Both fusion contrasts survive Holm correction.

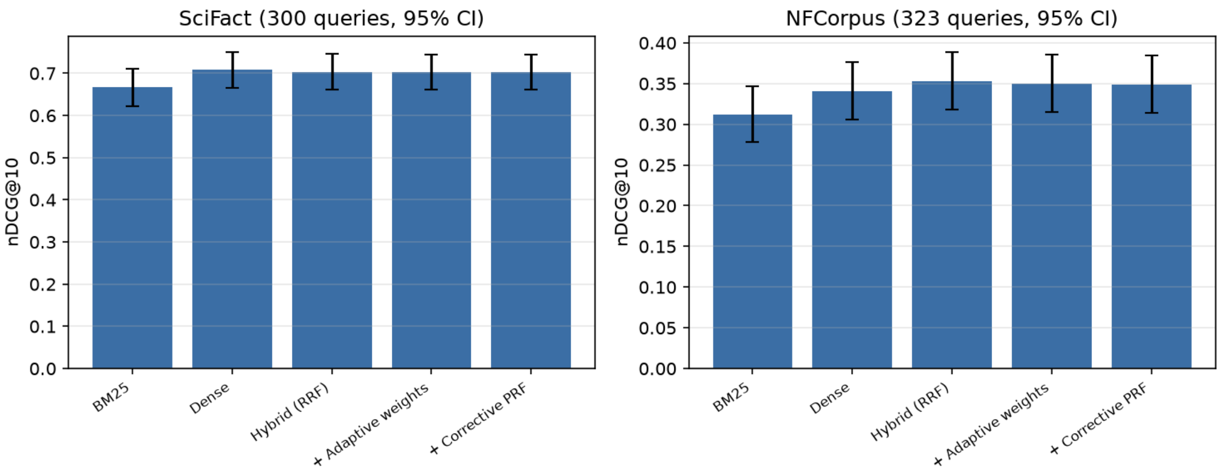

Figure 1: nDCG@10 across the ladder with 95% bootstrap confidence intervals.

Table 3: Cost vs. quality on SciFact (timed on 120 queries). The cross-encoder reranker is far slower and does not improve nDCG@10 over plain fusion; the corrective pass adds negligible cost.

System	nDCG@10	R@20	ms/query
Adaptive	0.734	0.900	28
Corrective PRF	0.734	0.900	43
Rerank (cross-enc)	0.702	0.886	2393

5.2 The corrective loop: quality at a fraction of the cost

Because BGE's top-1 similarity is high on nearly all SciFact claims, the confidence gate rarely fires—at most 6% of queries even at an aggressive threshold—so the corrective pass behaves almost identically to plain fusion at negligible extra cost (43 vs. 28 ms/query). The contrast with reranking is purely cost: the cross-encoder reranker is 87× slower (2393 ms/query) for no measurable nDCG@10 gain over plain fusion (0.702 vs. 0.734; Table 3, timed on the same 120 queries). Neither second pass earns its keep on these benchmarks: the reranker is far more expensive for nothing, and the corrective gate, where a strong embedder already retrieves confidently, simply does not fire—a genuinely negative result we report rather than dress up.

5.3 Faithfulness as evidence detection

We score every sentence of a claim's cited abstract for whether it is gold evidence, using NLI relatedness ($1 - P(\text{neutral})$), lexical overlap, and a relevance cross-encoder, and evaluate as threshold-free detection over 188 dev claims (gold base rate 0.18). Table 4 reports the result, and it is a useful corrective to the intuition that an entailment model is the obvious verifier: the relevance cross-encoder is the strongest detector (0.755 ROC-AUC, +0.067 over NLI), with a claim-bootstrap 95% CI that clears both baselines. NLI relatedness (0.688, 95% CI 0.652–0.726) is statistically indistinguishable from lexical overlap (0.686)—their intervals almost entirely overlap—so the entailment model buys nothing over word overlap on this task. The lesson is methodological as much as empirical—reported as detection against baselines that can lose, rather than against a citation check that scores zero by construction, a content-inspecting signal is a real, usable measure of support, and

Table 4: Evidence-sentence detection on SciFact dev. ROC-AUC (with a claim-bootstrap 95% CI) and PR-AUC are threshold-free; F1 is at a threshold tuned on a disjoint train split. Best in bold.

Scorer	ROC-AUC [95% CI]	PR-AUC	F1
Lexical overlap	0.686 [0.65, 0.72]	0.371	0.112
Cross-encoder	0.755 [0.72, 0.79]	0.476	0.469
NLI (DeBERTa)	0.688 [0.65, 0.73]	0.331	0.368

the cross-encoder already in the retrieval pipeline doubles as the best cheap verifier on this task. The ordering is unambiguous on the threshold-free ROC-AUC and PR-AUC; the F1 column uses a threshold tuned on a disjoint SciFact train split rather than on the dev set it scores, so it reflects operating-point performance without peeking. Turning the raw scores into a calibrated abstention gate is left to future work, since the three scorers' outputs are on different scales and not directly comparable as confidences.

6 Discussion

The practical lesson is an ordering of effort. For local scientific retrieval on a CPU budget, the first and largest win is a modern small embedder; reciprocal rank fusion adds a reliable, free improvement, especially on longer documents; and a cross-encoder reranker buys little for its one-to-two order of magnitude latency cost on these benchmarks. A cheap confidence-gated re-retrieval is a better place to spend a second pass than a reranker. For verification, the distinction that matters is content versus syntax: a content signal—a relevance cross-encoder, or an NLI model—reported honestly as a detector gives a real measure of whether a source supports a claim, where a citation check cannot.

7 Limitations

Both benchmarks are biomedical-leaning; broader transfer is untested. We use binary relevance, so NFCorpus numbers are not directly comparable to graded-nDCG leaderboard scores. The NLI verifier operates at the sentence level, so multi-sentence reasoning is only partially captured, and generation quality itself is not scored here—the end-to-end harness exists (nexusrag.eval.generation) but a small CPU generator is uninformative for it.

8 Conclusion

We turned a typical local RAG application into a measured one and found that the component everyone changes least—the embedder—matters most, while the components everyone advertises most—reranking and a corrective loop—are, at this scale, either not worth their cost or best made cheap. We hope the artifact is a useful template for the many similar systems that ship without numbers.

Reproducibility. pip install -e ".[eval]" then make eval, make corrective, and make faithfulness regenerate every table on CPU.

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